

Integrated bundled measures could halve reactive nitrogen losses and reach net-zero greenhouse gas emissions in global livestock systems

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Yubo Cao¹, Ling Liu², Tom Missellbrook³, Pete Smith⁴, Lin Ma⁵, Hao Zhao⁶, Xinpeng Jin⁷, Lu Zhang⁵, Gang Wang⁸, Keren Ding⁹, Xuan Wang⁸✉, Yakai Wang¹⁰, Juan Liu¹¹, Hongge Wang⁸, Zelong Liu¹², Atul Kumar¹³, Yongqing Guo¹⁴, Chenfeng Tian¹⁴, Zewen Tan¹ & Weimin Sun¹✉

Simultaneous mitigation of reactive nitrogen and greenhouse gas emissions in livestock systems is a critical challenge for sustainable food production. Here we conduct a meta-analysis of over 3,000 empirical observations to assess the mitigation performance of integrated technologies within livestock farms. Mitigation options are categorized into low-cost and easily scalable technologies and precision-integrated technologies, with the latter performing better by reducing reactive nitrogen and non-CO₂ greenhouse gas emissions by two-fifths and one-third, respectively. High-efficacy hotspots emerged in North America, Europe and East and Southeast Asia. Beyond current practices, a transition towards an integrated management framework centred on enclosed manure-treatment systems offers further reductions of over half for reactive nitrogen and two-thirds for non-CO₂ greenhouse gas by 2050. Meeting net-zero targets will require sustained capture of almost three-quarters of CO₂ from these systems. These findings highlight the critical role of integrated approaches for advancing sustainable and climate-resilient livestock production systems.

Livestock production underpins global economic development, rural livelihoods and nutritional security and contributes directly to several United Nations Sustainable Development Goals¹. However, emissions of reactive nitrogen (Nr) and non-CO₂ greenhouse gases (GHGs) pose substantial risks to air quality, human health and climate stability^{2,3}. Livestock and manure management currently account for an estimated 40–52% of anthropogenic ammonia (NH₃) emissions^{4,5} and 30–40% of global non-CO₂ GHG emissions, including methane (CH₄) and nitrous oxide (N₂O)^{5–7}. Without targeted intervention, these emissions will intensify with rising demand for animal-sourced foods⁸, underscoring the urgency of mitigation in the United Nations 2030 Agenda⁹.

Meeting global protein demand while reducing emissions remains a persistent challenge². Over the past four decades, numerous

mitigation technologies have been developed at laboratory and pilot scales⁵. Europe provides a reference case: coordinated policy and technological interventions, such as the Nitrates Directive (1991) and National Emission Ceilings Directive (2001), have led to reductions of more than 20% in NH₃ emissions and -16% in nitrogen (N) leaching^{4,10}. However, these advances have not translated into consistent global outcomes because existing assessments remain highly fragmented. Most studies focus on single pollutants (CH₄^{11,12}; N₂O¹³; NH₃¹⁴), individual manure-management practices^{15,16} or specific livestock species¹⁷. Such compartmentalized analyses limit system-level integration and hinder the identification of cross-pollutant trade-offs^{2,18}. Moreover, pronounced regional heterogeneity in livestock species, production systems and manure-management practices limits the

A full list of affiliations appears at the end of the paper. ✉e-mail: wx2008xw@yeah.net; wmsun@soil.gd.cn



Fig. 1 | Historical and spatial emissions of reactive nitrogen and greenhouse gases from global livestock farms. a, b, Historical trends in Nr (a) and GHG (b) emissions from livestock farms. **c, d**, Spatial distribution of Nr (c) and GHG (d) emissions from livestock farms in 2023. LU, livestock unit. Regional groupings are detailed in Supplementary Table 1.

transferability of mitigation strategies². These gaps underscore the need for an integrated framework capable of evaluating how interventions shape Nr and GHG emissions across diverse regional and management contexts.

To address these limitations, a global mitigation dataset was developed based on a meta-analysis of >3,000 empirical observations across four major livestock species—cattle, sheep/goats, chickens and pigs—and diverse manure-management systems. The dataset integrates direct and indirect non-CO₂ GHG indicators (enteric CH₄, manure CH₄, N₂O) and Nr pathways (NH₃, N₂O, NO_x and N leaching/runoff)¹⁸. To evaluate feasibility, mitigation measures were classified into two tiers according to cost-effectiveness and scalability^{11,14}: low-cost and easily scalable technologies (LCEST) and precision-integrated technologies (PIT). Although technical progress has been considerable, most applications remain confined to either enteric¹² or manure-related emissions^{13,14}, with limited coordination across the full emissions chain. To overcome these limitations, we propose an integrated, source-to-end framework centred on enclosed manure-treatment systems, offering an integrated mitigation solutions to support sustainable and climate-resilient transformation of global livestock systems.

Results

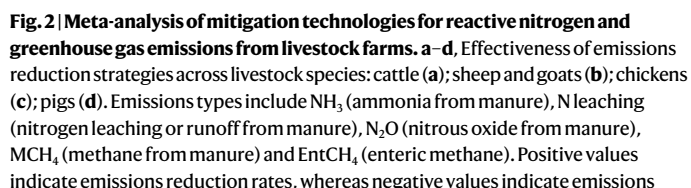
Efficiency gains to targeted mitigation address regional heterogeneity in livestock Nr and GHG emissions

Over the past four decades, productivity gains have substantially reduced the emissions intensity of livestock production. Between 1980 and 2020, Nr losses per kilogram of livestock protein declined from 0.94 kg to 0.59 kg, whereas GHG emissions per kilogram of livestock

protein dropped by 44%, from 76.0 kg CO₂-equivalent (CO₂eq) to 42.4 kg CO₂eq. However, per-animal emissions have remained largely unchanged (Fig. 1a,b). These patterns indicate that historical improvements largely reflect productivity-led efficiency gains¹⁹ rather than the adoption of explicit mitigation measures. Further reductions will therefore require a shift from efficiency-driven growth towards targeted, system-level mitigation.

Marked spatial heterogeneity underlies global livestock Nr losses (Fig. 1c). Grazing-based systems dominate in Oceania (>90% of regional losses), followed by Latin America and the Caribbean (74%) and Sub-Saharan Africa (61%). By contrast, confined manure systems are the major contributors elsewhere: in Europe, solid and liquid/slurry storage together account for half of total Nr losses, whereas in North America and East and Southeast Asia, solid storage is the leading source, followed by grazing systems. South Asia, Near East and North Africa and Sub-Saharan Africa exhibit mixed configurations, where drylot, grazing and solid storage jointly account for most losses. Species- and system-specific Nr loss patterns are provided in Supplementary Text 1.

Similar regional contrasts characterize GHG emissions from manure management (Fig. 1d). Although enteric CH₄ remains the dominant global source, the composition of manure-related emissions varies markedly across production systems. Grazing systems account for most manure-related emissions in Oceania (72%) and about half in Latin America and the Caribbean. By contrast, confined manure systems dominate in most other regions. In Europe, liquid/slurry and solid storage together comprise 70% of manure-related emissions, whereas lagoon, liquid/slurry and solid storage systems collectively contribute 67% and 58% in North America and East and Southeast Asia, respectively. In South Asia,



Near East and North Africa and Sub-Saharan Africa, solid storage and drylot remain the primary sources. Species- and system-specific GHG emissions patterns are detailed in Supplementary Text 2.

These spatial patterns link directly to differences in livestock species composition and manure-management structures, highlighting pronounced regional disparities in both Nr and GHG emissions.

Such heterogeneity indicates that uniform mitigation approaches are unlikely to be effective. Instead, strategies must be tailored to regional livestock-system structure to deliver meaningful reductions at scale.

Assessment and integration of existing mitigation technologies

Most existing interventions for livestock emissions control have concentrated on enteric CH₄¹² and NH₃ emissions from cattle and pigs^{5,15,16}, achieving notable pollutant-specific reductions. However, these single-pollutant approaches often lead to environmental trade-offs that limit their net effectiveness (Fig. 2). For example, feed additives such as ionophores and nitrates suppress enteric CH₄ formation but can elevate manure-derived N₂O emissions. Similar trade-offs arise from dietary interventions: high-fibre diets formulated to lower NH₃ volatilization in pig production can increase CH₄ emissions from manure storage by up to 28%. In manure management, simple covers substantially mitigate NH₃ emissions but promote aerobic surface nitrification²⁰, thereby increasing N₂O emissions. Although emerging options—such as acidification and cooling of liquid manure—can reduce NH₃, CH₄ and N₂O simultaneously, their uptake is constrained by high capital requirements and operational complexity^{21,22}.

To systematically assess mitigation options, we classified available measures into two functional tiers based on cost-effectiveness and deployment feasibility. The first tier (LCEST) includes interventions requiring limited capital or technical capacity, such as reduced dietary crude protein and manure-storage covers. A full list of LCEST measures is provided in Supplementary Table 2, and their cost-effective performance has been demonstrated in previous assessments^{11,14}. The second tier (PIT) comprises optimized combinations of currently available, region-specific best-performing measures tailored to local livestock species and manure-management systems (Supplementary Table 3). This integrated tier is designed to reduce both Nr and GHG emissions while minimizing cross-pollutant trade-offs. The conceptual architecture of this tiered approach is outlined in Supplementary Fig. 1. By aligning mitigation options with regional livestock-system structure and technological readiness, this tiered framework offers a scalable, system-oriented pathway for reducing Nr and GHG emissions in global livestock systems.

Potential and challenges for livestock Nr and GHG mitigation

Livestock farming contributes substantially to global environmental burdens, generating 46.3 Tg Nr and 3.9 Pg CO₂eq in 2023 (Fig. 3a,b), consistent with estimates from the Food and Agriculture Organization of the United Nations (FAO). Under LCEST, global livestock systems achieve reductions of 7.6 Tg Nr (16%) and 294 Tg CO₂eq (7.5%) (Fig. 3a,b), yet notable trade-offs persist; for example, simple manure covers can elevate N₂O emissions, partially offsetting gains (Fig. 3c,d). These findings illustrate the limitations of LCEST. By contrast, PIT delivers substantially deeper abatement—18.5 Tg Nr (40%) and 1,303 Tg CO₂eq (33%)—equivalent to ~10% of global anthropogenic Nr²³ and non-CO₂ GHGs^{8,24}.

Mitigation potential varies strongly across livestock species and management systems. In ruminant farms, grazing systems remain dominant sources of Nr losses, whereas confined-farm losses concentrate in solid storage and drylot systems. Limited availability of cost-effective technologies for grazing systems constrains mitigation under LCEST. Even within confined systems, LCEST achieves only modest reductions—26% for cattle and 19% for sheep/goats—due primarily to the lack of affordable drylot solutions (Fig. 3c). PIT yields greater reductions (33% and 29%, respectively), driven by large improvements in confined systems (47% for cattle; 66% for sheep/goats), including 50% mitigation in solid storage and 43% in drylot. In grazing systems, mitigation remains comparatively low at 19% (Fig. 3e and Extended Data Fig. 1). These patterns indicate that current technologies are markedly more effective in confined systems but poorly suited to diffuse emissions characteristic of extensive grazing operations.

Monogastric systems show higher overall mitigation rates (Extended Data Fig. 2) owing to their confinement-based nature. In chicken farms, manure with or without litter is the dominant Nr source, followed by pit storage and solid storage (Extended Data Fig. 1). LCEST reduces Nr losses by 19%, with manure with/without litter and solid storage accounting for 71% of total mitigation (Fig. 3c). PIT further enhances mitigation to 63%, with over 90% of reductions arising from manure with/without litter, pit storage and solid storage systems (Fig. 3e and Extended Data Fig. 2). In pig systems, liquid/slurry and solid storage account for roughly half of total Nr losses (Extended Data Fig. 1), and both frameworks achieve strong performance—55% under LCEST and 72% under PIT—enabled by high-efficiency (>50%) NH₃ abatement technologies (Fig. 3c,e).

GHG patterns mirror those for Nr. Enteric CH₄ remains the largest source (Extended Data Fig. 3), with reduction potentials of 236 Tg CO₂eq per year under LCEST and 697 Tg CO₂eq per year under PIT (Fig. 3d,f). Mitigation within grazing systems remains limited—11% for cattle and 9.9% for sheep/goats under LCEST and 14–15% under PIT—whereas confined systems yield substantially higher reductions (47% for cattle; 43% for sheep/goats under PIT). Solid manure management (including solid storage and drylot) remains the principal source of manure-related GHGs in confined ruminant systems, contributing 65% of cattle and 94% of sheep/goat emissions (Extended Data Fig. 3). PIT reduces these by 65%, primarily through N₂O abatement, whereas LCEST achieves only modest reductions (11% for cattle; 2.9% for sheep/goats). In monogastric systems, PIT again outperforms LCEST: chicken manure with/without litter, liquid/slurry and solid storage achieve reductions of 40%, 86% and 60%, respectively, whereas LCEST achieves minimal gains. In pig farms, lagoon, pit storage and slurry systems dominate CH₄ emissions (Extended Data Fig. 3), and low-cost covers frequently elevate N₂O, limiting LCEST effectiveness (<3%), whereas PIT achieves reductions of 195 Tg CO₂eq per year (72%), with >80% mitigation across key subsystems (Fig. 3d,f).

Overall, these findings highlight the substantially greater mitigation achievable through integrated, higher-efficiency technologies within existing management systems. Performance is markedly higher in monogastric systems owing to strong compatibility with confinement-based technologies, whereas persistent underperformance in grazing systems reflects structural and technological limitations that continue to constrain large-scale mitigation in extensive production systems.

Regionally divergent mitigation outcomes call for differentiated strategies and targeted innovation in livestock systems

Mitigation performance varies markedly across regions, reflecting differences in livestock-system structure and technological readiness. Mitigation effectiveness increases with the degree of confinement for Nr and non-CO₂ GHGs, but this relationship is absent for GHGs under the LCEST framework (Fig. 4a–d). Under LCEST, GHG reductions remain below 10% even in high-confinement systems, underscoring persistent gaps in cost-effective CH₄ and N₂O abatement technologies. Regions dominated by confined systems—North America, Europe and East and Southeast Asia—achieve the highest mitigation rates, with Nr reductions exceeding 29% under LCEST and 46% under PIT and GHG reductions under PIT surpassing 43%. These outcomes reflect long-standing investments in research and development tailored to local manure-management structures.

By contrast, mitigation performance remains comparatively weak in South Asia, Latin America and the Caribbean, Oceania, Near East and North Africa and Sub-Saharan Africa (Fig. 4a–d). Constraints stem from limited access to regionally adapted technologies (South Asia, Near East and North Africa, Sub-Saharan Africa) and the prevalence of grazing-based systems (Latin America and the Caribbean, Oceania) (Fig. 1c,d), despite promising results from confined-system trials (Extended Data Fig. 4). Countries combining high-emissions intensity

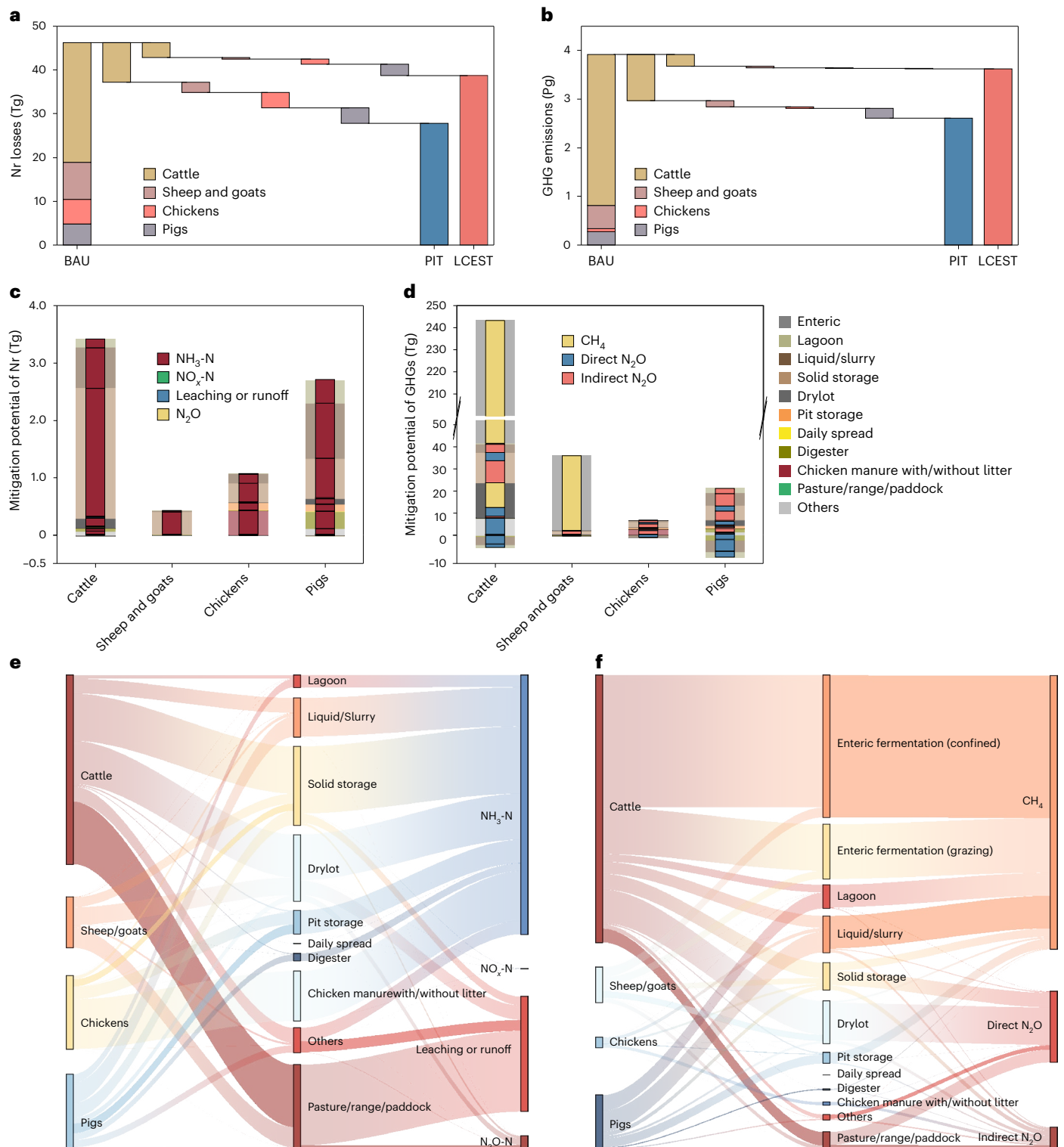


Fig. 3 | Mitigation potential of reactive nitrogen and greenhouse gas emissions under different mitigation scenarios. Comparative impacts of low-cost, scalable technologies (LCEST) and precision-integrated technologies (PIT) on emission reductions. **a, b**, Nr losses (**a**) and GHG emissions (**b**) under a business-as-usual (BAU) scenario and two mitigation scenarios. **c, d**, Mitigation potentials for Nr losses (**c**) and GHG emissions (**d**) achieved through LCEST. **e, f**, Distribution of mitigation contributions for Nr losses (**e**) and GHG emissions (**f**) across livestock species, management systems and emissions pathways

under the PIT scenario. Sankey diagrams illustrate the flow of total mitigation from livestock species (left nodes) through management systems (middle nodes) to emissions pathways (right nodes). Flows from livestock species nodes show how mitigation is distributed across management systems, whereas flows from management-system nodes to emissions-pathway nodes indicate which pollutants contribute to the achieved mitigation. Flow width represents the magnitude of emissions reductions. Mitigation technologies and assumed reduction rates are summarized in Supplementary Table 4.

with large livestock populations—India, Brazil, Ethiopia and Pakistan—could hold substantial mitigation potential (Extended Data Fig. 5), yet realized reductions remain modest (Extended Data Fig. 6) owing to the

scarcity of technologies suited to locally dominant manure systems such as drylot. These disparities highlight the need for efficient, locally adaptable mitigation options.

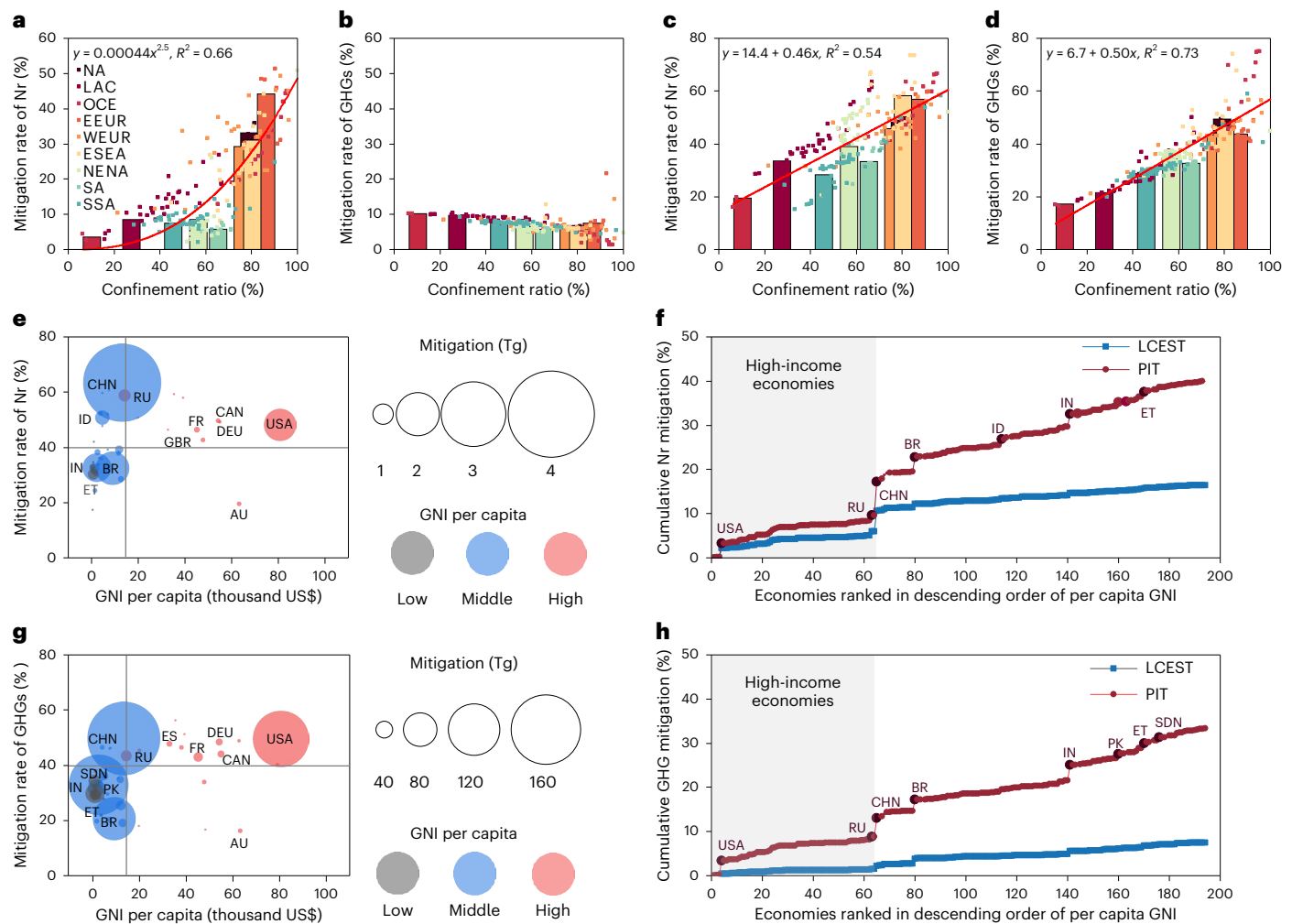


Fig. 4 | Spatial variations in reactive nitrogen and greenhouse gas emissions reduction rates under different mitigation scenarios. **a,b**, Relationships between confinement ratio and mitigation rate of Nr (**a**) and GHGs (**b**) under the LCEST scenario. Each dot represents national-level data. **c,d**, Relationships between confinement ratio and mitigation rate of Nr (**c**) and GHGs (**d**) under the PIT scenario. **e,g**, Mitigation performance by country income level under the PIT scenario for Nr (**e**) and GHGs (**g**). Countries are classified into four quadrants based on income and emissions reduction performance: high-income, high-performing (above 40%) countries; high-income, low-performing countries; low-/middle-income, high-performing countries and low-/middle-income, low-performing countries. Details in

Supplementary Tables 5 and 6. **f,h**, Contributions of economies to global mitigation of Nr (**f**) and GHGs (**h**) under different scenarios, ranked by gross national income (GNI) per capita (Atlas method). GNI data from the World Bank (2023). Countries marked are the top emissions reduction contributors. AU, Australia; BR, Brazil; CAN, Canada; CHN, China; DEU, Germany; EEUR, Eastern Europe; ES, Spain; ESEA, East and Southeast Asia; ET, Ethiopia; FR, France; GBR, United Kingdom; ID, Indonesia; IN, India; LAC, Latin America and the Caribbean; NA, North America; NENA, Near East and North Africa; OCE, Oceania; PK, Pakistan; RU, Russia; SA, South Asia; SDN, Sudan; SSA, Sub-Saharan Africa; WEUR, Western Europe.

Given this heterogeneity, mitigation strategies must align with national capacities and technological readiness rather than assume a uniform global pathway. Accordingly, countries were grouped into four typologies by income level and achieved mitigation performance (Fig. 4e,g). High-income, high-performing countries—such as the USA, Canada, France and Germany—are well positioned to lead early transitions through established infrastructure and regulatory frameworks. Australia, despite its high-income status, illustrates the limits of this group, as its predominantly extensive systems require context-specific innovation. Restricting PIT implementation to high-income regions yields only 9.6% and 8.7% reductions in global Nr and GHG emissions, respectively. By contrast, targeting high-emissions-intensity countries provides far greater leverage: deploying PIT in only 7–8 such countries could generate nearly half of global abatement (Fig. 4f,h). This concentration of potential highlights the disproportionate influence of a small number of countries on global emissions trajectories.

Enabling deep emissions reductions through the 3C mitigation framework

Existing mitigation strategies target enteric CH_4 and manure emissions but overlook upstream factors (for example, feeding regimes and animal traits) and downstream sources (for example, post-treatment exhaust gases). The 3C framework addresses these gaps through coordinated interventions to cut emissions at the source, curb formation during processing and capture residual gases post-treatment (Fig. 5; scenario details provided in the Methods). Implemented globally, the framework substantially enhances mitigation outcomes. Even without carbon capture and sequestration (CCS), livestock-farm Nr and GHG emissions decline by 58% and 69%, respectively, by 2050 (Fig. 6a,b). Major regions—including North America, Europe, East and Southeast Asia, South Asia and the Near East and North Africa—achieve Nr reductions exceeding 60% and GHG reductions exceeding 70% (Fig. 6c and Extended Data Fig. 7). With CCS integration, farms in these high-performing regions could, in principle, achieve net-negative

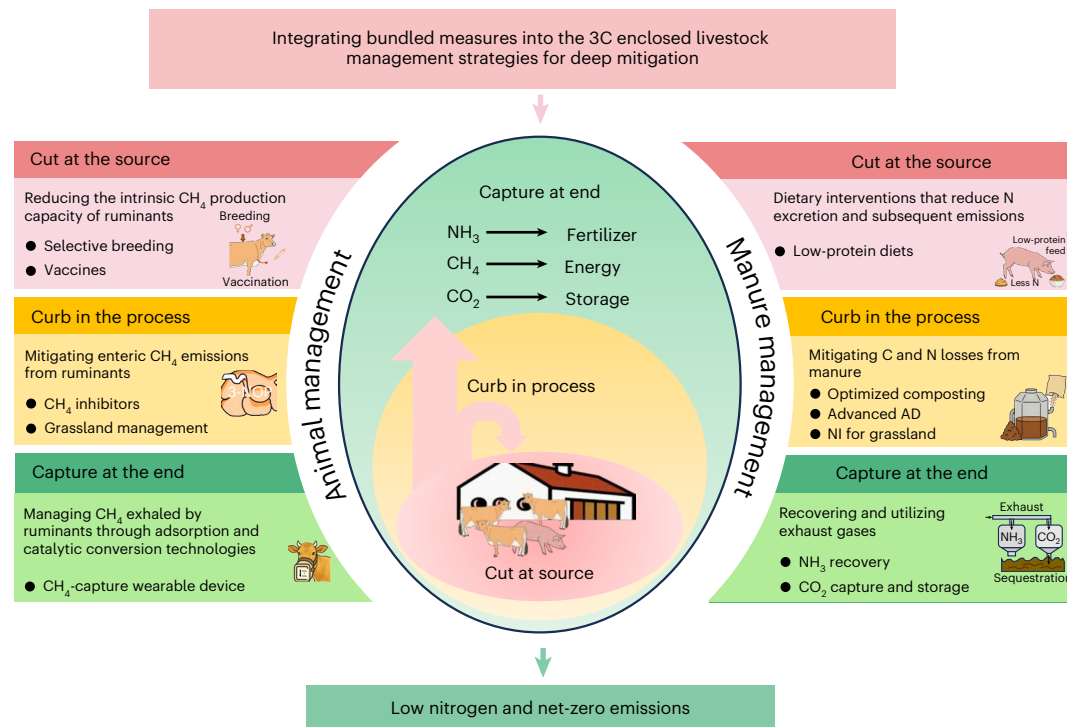


Fig. 5 | Conceptual framework of the holistic source-to-end management strategy. The framework is built on the 3C approach—cut at the source, curb in the process and capture at the end—designed to develop enclosed livestock-management systems that minimize Nr losses and achieve net-zero GHG emissions. This approach targets the full emissions chain, encompassing

intrinsic emissions potential, biological transformation processes and final exhausts, enabling holistic source-to-end mitigation across livestock systems. Detailed information regarding the integration of these strategies within the 3C framework is provided in Supplementary Table 7. Icons adapted with permission from BioGDP.com (ref. 50). AD, anaerobic digestion; NI, nitrification inhibitor.

emissions (Fig. 6d). This outcome is technically plausible because projected regional geological CO₂ storage capacity (~16 Pg CO₂ per year by 2050)²⁵ far exceeds the residual emissions remaining under the 3C pathway (1.4 Pg CO₂eq per year; Fig. 6b), although only a fraction of this capacity would be accessible to the livestock sector.

Discussion

Implications for deploying integrated technologies in existing livestock systems

Rising demand for animal-sourced foods continues to intensify pressure on mitigation efforts^{2,8}. Historical improvements in livestock productivity have reduced emissions intensities (Fig. 1a,b), but growing evidence indicates that this trend is plateauing¹⁹, highlighting the need for explicit mitigation actions. To date, many interventions have been fragmented—focusing on single pollutants^{11–14} or isolated technologies^{5,15,16}—and have therefore overlooked interactions across pollutants, management structures and unintended trade-offs. Approaches that integrate pollutant linkages with regional system characteristics offer a more comprehensive basis for mitigation. Crucially, such integrated strategies can reduce emissions without compromising productivity. Manure-management technologies generally avoid production penalties⁵, and feeding interventions—such as reduced-protein diets and 3-nitrooxypropanol—can maintain productivity when appropriately implemented, although long-term responses remain insufficiently understood¹².

Beyond direct environmental gains, integrated mitigation strategies offer substantial environmental and societal benefits. Under the LCEST and PIT scenarios, the estimated global benefits value of combined Nr and GHG reductions reaches US\$16.9 billion per year and US\$38.6 billion per year, respectively (Supplementary Fig. 2). Improved manure management also enhances N retention, which strengthens nutrient circularity and decreases dependence on synthetic fertilizers, particularly in nutrient-deficient regions such as Sub-Saharan Africa²⁶

and decreases nitrogen surpluses in high-input systems (for example, China and the USA)⁴, provided that recovered manure nutrients are effectively managed and spatially matched to cropland²⁷.

Mitigation technologies in livestock systems face substantial barriers to implementation, particularly in low- and middle-income countries (LMICs)²⁸. These regions account for more than 70% of the global mitigation potential quantified in this study (Fig. 4f,h). Within LMICs, high-productivity systems provide the most immediate and scalable entry point for intervention: deploying PIT in such systems could deliver 47% of total Nr and 39% of total GHG reduction potential across LMICs (Supplementary Fig. 3). China provides an illustrative case: between 2007 and 2017, >US\$770 million in government investment enabled the establishment of 5,000 large-scale farms with upgraded manure-management infrastructure, substantially reducing Nr losses²⁹. Complementary Europe–China technology transfer further accelerated adoption of advanced mitigation strategies such as amino-acid-balanced low-protein diets³⁰. Economic assessments show that the societal benefits of Nr and GHG reductions exceed abatement costs^{11,29}. Despite these successes, high up-front costs constrain broader adoption across LMICs. Evidence from the International Fund for Agricultural Development (IFAD)-supported projects shows that international finance can generate measurable GHG reductions—particularly methane—relative to non-investment baselines³¹. Unlocking LMIC mitigation potential will therefore require coordinated support across regulatory incentives, technology transfer and sustained financial assistance.

Towards holistic mitigation in livestock systems through the 3C framework

Current mitigation outcomes reveal fundamental technical constraints in prevailing livestock and manure-management systems. In this context, the 3C framework provides the broader system transformation required for the livestock sector to align with the 1.5 °C

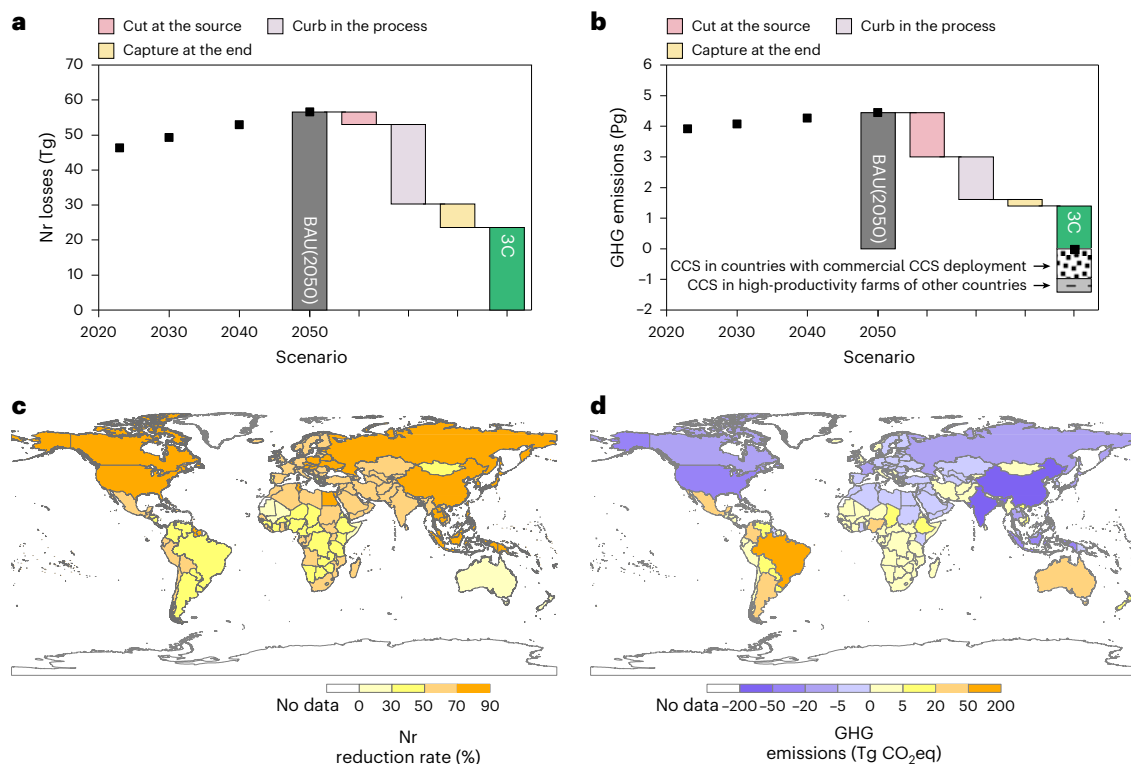


Fig. 6 | Projected mitigation outcomes under different scenarios.

a,b, Projected emissions of Nr (**a**) and GHGs (**b**) under different scenarios, illustrating the phased reduction potential of the 3C framework by 2050. **c,d**, Regional variation in Nr reduction rates (**c**) and corresponding GHG emissions (**d**) under the 3C enclosed livestock-management framework by 2050. In the 3C scenario, countries operating commercial carbon-capture-and-storage projects, together with high-productivity confined farms in other regions, are assumed to

achieve 90% capture and permanent storage of available CO₂. ‘Available CO₂’ refers to exhaust emissions from enclosed manure-management systems—including advanced anaerobic digesters and sealed composting units. Negative values in panel **d** indicate that manure-derived CO₂ captured and sequestered exceeds total farm-level GHG emissions. Basemap administrative boundary data in **c** and **d** from the Resource and Environment Science Data Platform, Chinese Academy of Sciences (<https://www.resdc.cn/data.aspx?DATAID=205>).

target³². Central to the 3C pathway is the transition of conventional manure handling towards enclosed management systems. Without physical containment, post-processing emissions cannot be consistently controlled. Enclosed anaerobic digestion for liquid manure and in-vessel composting for solids enable both deep reductions in Nr and GHG emissions while allowing residual gases to be captured for downstream treatment^{33,34}. Evidence from China illustrates that large-scale transitions towards enclosed manure-management systems are technically and operationally feasible when supported by strong policy and financial incentives¹³. By 2020, more than 100 large-scale livestock farms had adopted enclosed in situ composting reactors for solid-manure treatment³⁵, including in less economically developed regions (Supplementary Fig. 4a). Some installations also incorporated exhaust-gas recovery³³. Advanced anaerobic digestion for liquid manure is even more widespread, accounting for 42% of treatment among large-scale farms³⁵. In parallel, centralized manure-treatment hubs serving smallholders—publicly financed and enterprise-operated—have been successfully established³⁶ (Supplementary Fig. 4b), demonstrating feasibility in regions with limited technical and capital capacity. Together, these interventions contribute to measurable reductions in local emissions^{13,36}, whereas revenues from biogas and organic fertilizers can partially offset investment and operating costs^{33,37}. Nevertheless, full implementation of the 3C framework requires substantial up-front capital and may be constrained in LMICs without adequate economic incentives³⁸. Emerging financial instruments—including targeted credit schemes, premium-payment programmes, carbon-market mechanisms and results-based funding (Supplementary Table 8)—offer promising pathways to narrow this implementation gap.

Achieving net-zero emissions in the global livestock sector will also depend on expanding CCS infrastructure. Approximately 70% of CO₂ emissions from manure-treatment exhausts would need to be captured to reach net-zero outcomes (Extended Data Fig. 7a). Although more than 50 countries currently operate commercial CCS facilities (Supplementary Table 1), restricting deployment to farms within these CCS-active regions would offset only 69% of livestock-derived GHGs. This demonstrates that the current global CCS infrastructure is insufficient to support net-zero outcomes in livestock systems and that substantial scaling of CCS capacity alongside strengthened international cooperation and technology transfer will be required. Examples such as Japan’s support for CCS deployment in Malaysia and Indonesia (Supplementary Table 1) demonstrate how cross-border partnerships accelerate the diffusion of gas-separation technologies and geological-storage infrastructure.

High-productivity livestock farms, characterized by industrial-scale production and concentrated waste streams (Supplementary Text 3), represent the most feasible entry points for CCS deployment. Extending CCS to high-productivity farms outside CCS-active regions could make the net-zero outcome in Fig. 6b attainable, assuming CCS efficiencies approach 90% (ref. 32), consistent with the upper end of performance reported for manure-treatment facilities operating under favourable geological and operational conditions^{34,39,40}. Importantly, this efficiency represents a technical upper bound rather than a near-term universal expectation²⁵. Realizing such performance at scale will depend on technological readiness and strengthened national climate ambition and enabling policy and financial environments.

Whereas the proposed 3C framework provides a technically plausible pathway towards low-emission livestock systems, it remains

conceptual at this stage. Many individual technologies have been validated under pilot- or farm-scale conditions (Supplementary Table 7), and some—such as anaerobic-digestion-based CCS and ammonia recovery from composting exhaust—are already being partially deployed in practice^{34,36}. However, no existing system integrates source-, process- and end-of-pipe interventions concurrently under real-world conditions. This reflects both the tight engineering coupling required among biological and gas-handling processes and the absence of field-scale pilots capable of evaluating combined, multi-stage interventions. Moving towards integrated deployment will require: (1) ensuring technical and operational compatibility among components; (2) maintaining process stability when multiple interventions are combined; and (3) establishing robust, long-term evaluation frameworks capable of detecting interactive or emergent effects.

Finally, although our assessment captures synergies and trade-offs at the farm scale, mitigation actions inevitably interact with broader systems. Low-protein feeding can reduce land-use pressures through partial substitution of soybean meal⁴¹, whereas anaerobic digestion and composting generate biogas that displaces fossil fuels and recycle nutrients that reduce upstream emissions from mineral-fertilizer production^{42,43}. These cross-sectoral impacts were not quantified owing to limited region-specific data and difficulties in defining consistent system boundaries. Future assessments will benefit from incorporating local CCS capacity, governance and financing conditions⁴⁴ and projected technological advances in manure management and enteric methane mitigation, all of which may reshape achievable reductions towards mid-century. Nevertheless, as the first integrated global assessment of livestock-farm Nr and GHG mitigation potential, this study provides a foundation for continued refinement of mitigation pathways and informs the development of coordinated, system-level climate strategies for the livestock sector.

Methods

Global livestock mitigation technology database

A global dataset of mitigation outcomes for Nr and GHG emissions in livestock systems was compiled through a systematic meta-analysis of peer-reviewed studies published from 1989 to the present. Relevant studies were identified through systematic searches of Web of Science and Google Scholar. Keyword combinations included ‘ammonia OR greenhouse gas OR nitrous oxide OR methane OR leaching OR runoff OR enteric methane’, ‘mitigation OR reduction’ and ‘livestock farm’. Studies were included only when they (1) reported quantitative measurements of NH₃, N₂O, CH₄ or N leaching/runoff; (2) contained both treatment and control groups; and (3) focused on cattle, pigs, chickens or sheep/goats, which together account for the vast majority of global livestock emissions. For publications presenting data exclusively in graphical format, numerical values were extracted using GetData Graph Digitizer (version 2.22).

A total of 1,015 studies, comprising 3,334 treatment–control pairs, satisfied the inclusion criteria. These studies covered mitigation interventions across five major stages of livestock production: feeding, housing, manure storage/spreading, manure treatment and grazing. Emissions pathways evaluated included enteric CH₄, manure-derived CH₄, N₂O, NH₃ emissions and nitrogen leaching or runoff from manure. Across all records, 59 distinct mitigation technologies were identified. A detailed database of these technologies, together with their classification and performance metrics, is provided in Supplementary Table 9.

The effect size for each pair was computed as the natural logarithm of the response ratio (R), as elucidated in equation (1)⁴⁵:

$$\ln R = \ln \left(\frac{E}{C} \right) \quad (1)$$

where E represents the mean Nr or GHG emission for an experimental treatment, C denotes the mean Nr or GHG emission for the control

treatment. Because many studies did not report any measure of variance for the response variables, observations were weighted based on the replication number to include as many studies as possible, following equation (2)⁷:

$$\text{weight} = \frac{n_t n_c}{n_t + n_c} \quad (2)$$

where n_t and n_c are the sample sizes for the treatment and the control groups, respectively. Mean effect sizes and bias-corrected 95% confidence intervals were derived through a bootstrapping procedure comprising 5,000 iterations, utilizing Metawin 2.1.4 software. The results of the analyses for $\ln R$ were back-transformed and reported as percentage change in Nr or GHG emissions.

Estimation of livestock Nr and non-CO₂ GHG emissions

This study quantified both emissions and mitigation potential from on-farm sources, including enteric CH₄, manure management and excreta deposited during grazing. Emissions associated with manure applied to croplands—including arable land, forage crops (for example, corn silage) and grasslands for silage production—were excluded. These field emissions fall outside the farm-gate boundary adopted here, as they are primarily governed by soil–crop management practices rather than livestock-system interventions⁴⁰.

Reactive N losses (Q_{Nr}) were estimated using a source-based approach, incorporating livestock population data, manure-management systems, nitrogen excretion rates and corresponding emissions factors for NH₃-N, NO_x-N, N₂O-N and leaching/runoff:

$$Q_{Nr} = \sum_s \left[\sum_T (A_{(T)} \text{Nex}_{(T)} \text{MMS}_{(T,S)} \text{EF}_{Nr(T,S)}) \right] \quad (3)$$

where, $A_{(T)}$ denotes the number of livestock species T (cattle, pigs, chickens and sheep/goats), with historical data sourced from FAO (2023). Country-specific projections of livestock numbers from the present to 2050 were obtained using the Global Biosphere Management Model (GLOBIOM)⁴⁶. In GLOBIOM, livestock populations are endogenously determined based on regional demand for meat, milk and eggs, which evolves with income growth, dietary transitions and population dynamics under the Shared Socioeconomic Pathway 2 (SSP2) scenario. Annual N excretion rates ($\text{Nex}_{(T)}$) for each livestock species and productivity level (high- and low-productivity systems) were primarily derived from the Intergovernmental Panel on Climate Change (IPCC, 2019)⁴⁰. Country-specific values for major producers were further calibrated using national GHG inventories submitted to the United Nations Framework Convention on Climate Change (UNFCCC, 2022)⁴⁷ and peer-reviewed literature (Supplementary Table 10). $\text{MMS}_{(T,S)}$ denotes the share of manure excreted by species T that is managed under manure-management system S . Given the pronounced differences in manure-management practices between high- and low-productivity systems, system-specific shares were primarily derived from IPCC (2019)⁴⁰, with country-level adjustments applied where national inventory submissions⁴⁷ or supplementary literature were available (Supplementary Table 10). $\text{MMS}_{(T,S)}$ was allowed to vary over time to represent projected transitions in livestock production systems, as parameterized in Supplementary Table 11. Emissions factors $\text{EF}_{Nr(T,S)}$ for individual nitrogen pathways (leaching/runoff, N₂O, NH₃ and NO_x) were sourced from IPCC (2019)⁴⁰ and the Global Livestock Environmental Assessment Model (GLEAM)⁴⁸. In countries with widespread adoption of mitigation technologies—such as slurry acidification or manure-storage covers, particularly in Europe and North America—emissions factors were adjusted according to documented implementation rates (Supplementary Text 4).

Methane emissions (Q_{CH_4} , kg) were estimated as the sum of enteric fermentation ($Q_{\text{CH}_4,\text{enteric}}$, kg) and manure emissions ($Q_{\text{CH}_4,\text{manure}}$, kg):

$$Q_{\text{CH}_4} = Q_{\text{CH}_4,\text{enteric}} + Q_{\text{CH}_4,\text{manure}} \quad (4)$$

$$Q_{\text{CH}_4,\text{enteric}} = \sum_T A_{(T)} \text{EF}_{\text{CH}_4,\text{enteric}(T)} \quad (5)$$

$$Q_{\text{CH}_4,\text{manure}} = \sum_T A_{(T)} \text{EF}_{\text{CH}_4,\text{manure}(T)} \quad (6)$$

where $Q_{\text{CH}_4,\text{enteric}}$ and $Q_{\text{CH}_4,\text{manure}}$ (kg CH_4 per year) represent CH_4 emissions from enteric fermentation and manure management, respectively. $A_{(T)}$ is the number of livestock species T , and $\text{EF}_{\text{CH}_4,\text{enteric}(T)}$ and $\text{EF}_{\text{CH}_4,\text{manure}(T)}$ are species-specific emission factors (kg CH_4 per head per year). Values for $\text{EF}_{\text{CH}_4,\text{enteric}}$ were compiled from the IPCC 2019 Refinement⁴⁰ and national data from major emitting countries, as reported in the National GHG Inventories submitted to the UNFCCC (2022)⁴⁷ and peer-reviewed literature (Supplementary Table 10). Manure CH_4 emission factors ($\text{EF}_{\text{CH}_4,\text{manure}}$) were estimated using the IPCC Tier 2 methodology, with further details provided in Supplementary Text 5.

Total non- CO_2 GHG emissions (GHGE) were estimated by integrating methane and nitrogen-based emission pathways, following the equation:

$$\text{GHGE} = Q_{\text{CH}_4} \times 27 + Q_{\text{N}_2\text{O}} \times \frac{44}{28} \times 273 + Q_{\text{indirect}} \times E \times \frac{44}{28} \times 273 \quad (7)$$

Where Q_{CH_4} is total CH_4 emissions (kg per year), as defined in equation (4); $Q_{\text{N}_2\text{O}}$ is direct N_2O -N losses (kg N per year); Q_{indirect} accounts for nitrogen losses through indirect N_2O pathways, including NH_3 and NO_x deposition and leaching/runoff (kg N per year); E is the emissions factor for indirect N_2O -N formation, set at 0.014 kg N_2O -N per kg N for NH_3/NO_x deposition in wet climates, 0.005 kg N_2O -N per kg N in dry climates and 0.011 kg N_2O -N per kg leached N, following IPCC 2019 guidelines⁴⁰; 27 and 273 are the 100-year global warming potentials of CH_4 and N_2O , respectively, as recommended by the sixth IPCC Assessment Report³¹; The factor 44/28 accounts for the conversion of N_2O -N to N_2O mass.

Scenario design and mitigation strategy framework

Four scenarios were defined for this study: a business-as-usual scenario and three mitigation scenarios, including LCEST, PIT and the 3C enclosed livestock-management strategy. The business-as-usual scenario reflects a continuation of current practices with no additional mitigation measures. The LCEST scenario incorporated mitigation technologies that are widely available and economically feasible, as listed in Supplementary Table 2. To account for regional heterogeneity, these technologies were matched to emissions sources in each region (Supplementary Table 3), prioritizing measures that deliver co-benefits for both Nr and GHG mitigation. These region-specific packages formed the basis for the PIT scenario, which represents a practical integration of currently deployable, best-performing technologies without altering the existing manure-management structure. The methodological workflow is summarized in Supplementary Fig. 1. Emissions under each scenario were estimated by multiplying region- and source-specific activity data by corresponding emissions factors, adjusted to reflect the application of mitigation technologies.

The 3C scenario is a forward-looking framework designed to achieve deep Nr abatement and progress towards net-zero GHG emissions through coordinated source-, process- and end-of-pipe interventions (Fig. 5 and Supplementary Table 12). For enteric CH_4 , the 3C approach follows a Source–Process–End structure. Source-level interventions include selective breeding for low-emission ruminants and vaccine development targeting methanogenic archaea. Process-level measures suppress CH_4 formation using 3-nitrooxypropanol in confined systems and tanniferous forages in grazing systems. End-of-pipe

options involve CH_4 -capture devices—such as adsorptive materials or catalytic masks—that intercept exhaled methane before atmospheric release. All enteric methane mitigation technologies incorporated into the 3C scenario and their respective mitigation efficiencies are listed in Supplementary Table 7.

A parallel set of interventions was applied to manure management. Source-level measures reduce nitrogen excretion through low-protein, amino-acid-balanced diets. Process-level treatments differ by manure type: liquid manure is treated using advanced anaerobic digestion systems assuming low leakage and gastight storage, consistent with IPCC (2019) guidelines⁴⁰; solid manure is treated via optimized composting protocols that account for feedstock characteristics, aeration and the use of emission-reducing additives identified in the meta-analysis. Manure-type assignments are provided in Supplementary Table 13. In grazing systems, nitrification inhibitors were applied to control nitrogen losses from excreted waste. End-of-pipe strategies comprise exhaust-gas capture from enclosed manure-management systems. Ammonia recovery is applied only to exhaust air from solid-manure composting, where NH_3 concentrations are sufficiently high for effective gas-phase capture³³. By contrast, slurry-based anaerobic digestion generates much lower NH_3 concentrations^{42,43}, rendering gas-phase recovery inefficient and therefore not implemented. In both composting reactors and anaerobic digesters, CO_2 in the exhaust stream is captured and permanently sequestered. All manure-management technologies incorporated into the 3C scenario and their respective mitigation efficiencies are listed in Supplementary Table 7, with full accounting of CO_2 fluxes detailed in Supplementary Text 6. Technology adoption was parameterized based on technology-specific applicability and system compatibility. In the mitigation scenario, technologies were assumed to be fully implemented only within technically and contextually suitable systems (as detailed in Supplementary Tables 2, 12 and 14), thereby representing the technically achievable mitigation potential rather than universal adoption.

To evaluate synergistic mitigation benefits, we quantified the societal benefits of global livestock emissions reductions, defined as the aggregate of avoided ecosystem damages, health benefits and climate impacts (Supplementary Text 7). Region-specific evaluation procedures followed established assessment frameworks⁴⁹. Historical emission intensities of Nr and GHGs from 1980 to 2020 were derived by dividing total emissions by livestock protein production or livestock-unit equivalents using FAOSTAT data. Emissions included enteric fermentation, manure management and manure deposition on pastures; detailed methods are provided in Supplementary Text 8.

Statistics and reproducibility

This study is based on a meta-analysis of published empirical data and scenario-based modelling; no new experimental interventions were conducted. Sample sizes were determined by the availability of eligible studies identified through systematic literature searches. No statistical method was used to predetermine sample size. Data inclusion and exclusion criteria were defined a priori (Methods), and no data were excluded after extraction. As this study does not involve experimental treatments or subject allocation, randomization and blinding are not applicable. Statistical analyses were conducted using non-parametric bootstrapping procedures as described above. Sensitivity analyses were performed to evaluate the robustness of model outputs to parameter uncertainty.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The data supporting the findings of this study are available within the article and the Supplementary Information. Detailed data sources

used for the meta-analysis are listed in Supplementary Tables 9 and 15. The coefficients used in the mitigation-effect analysis are provided in Supplementary Tables 2, 4 and 7. Source data are provided with this paper.

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Author contributions

Y.C. and L.L. conceived and designed the study. Y.C. conducted the research and drafted the manuscript. L.L., T.M., P.S., X.J. and L.Z. critically revised the manuscript. H.Z. assisted with model simulations. Y.W. performed the sensitivity analysis. G.W. prepared the distribution maps. All authors contributed to the discussion and revision of the paper.

Competing interests

The authors declare no competing interests.

Additional information

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Correspondence and requests for materials should be addressed to Xuan Wang or Weimin Sun.

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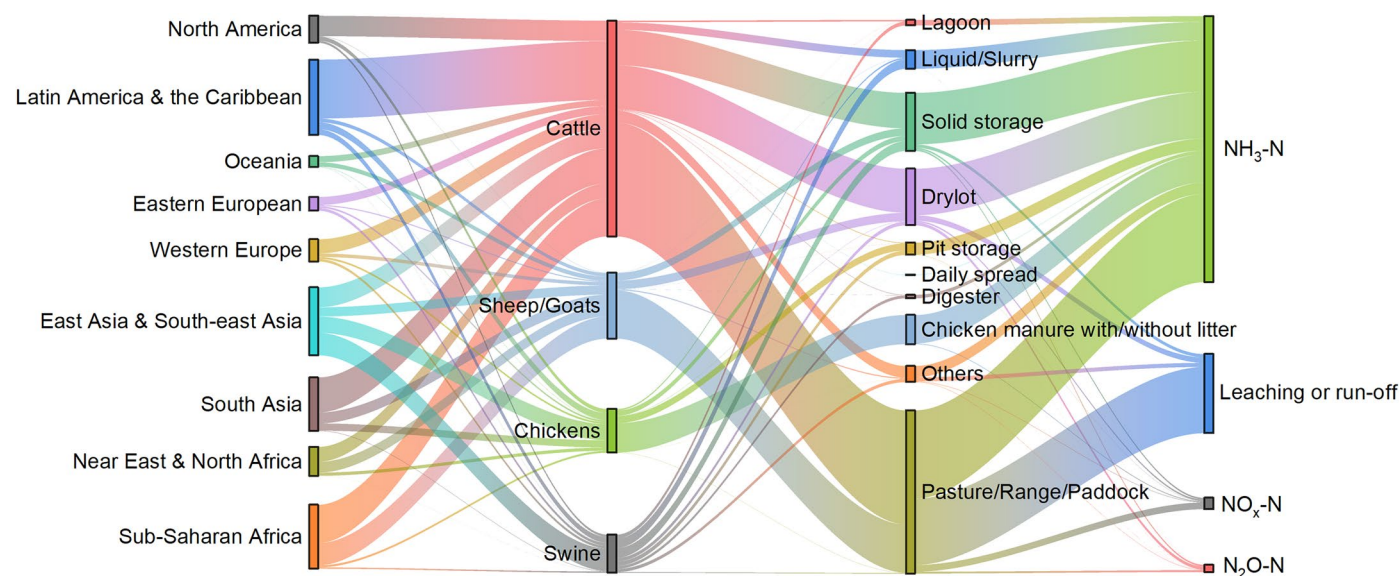
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¹National-Regional Joint Engineering Research Center for Soil Pollution Control and Remediation in South China, Guangdong Key Laboratory of Integrated Agro-environmental Pollution Control and Management, Institute of Eco-environmental and Soil Sciences, Guangdong Academy of Sciences, Guangzhou, China. ²College of Agronomy, State Key Laboratory of Crop Biology, National Engineering Research Center for Slow/Controlled Release Fertilizers, Shandong Agricultural University, Tai'an, China. ³Net-zero and Resilient Farming, Rothamsted Research, North Wyke, UK. ⁴Institute of Biological and Environmental Sciences, School of Biological Sciences, University of Aberdeen, Aberdeen, UK. ⁵State Key Laboratory of Pollution Control and Resource Reuse, School of the Environment, Nanjing University, Nanjing, China. ⁶Jiangsu Key Laboratory of Soil and Water Processes in Watershed, College of Geography and Remote Sensing, Hohai University, Nanjing, China. ⁷Institute of Environmental Sciences (CML), Faculty of Science, Leiden University, Leiden, The Netherlands. ⁸Key Laboratory of Agricultural Water Resources, Hebei Key Laboratory of Soil Ecology, Center for Agricultural Resources Research, Institute of Genetic and Developmental Biology, Chinese Academy of Sciences, Shijiazhuang, China. ⁹AgResearch, Ruakura Research Centre, Hamilton, New Zealand. ¹⁰Department of Natural Resources and the Environment, College of Agriculture, Health and Natural Resources, University of Connecticut, Storrs, CT, USA. ¹¹Soil and Physical Science Department, Lincoln University, Christchurch, New Zealand. ¹²School of Agriculture, Food and Ecosystem Sciences, Faculty of Science, The University of Melbourne, Parkville, Victoria, Australia. ¹³IPCA Centre for Waste Management and Research, TERI School of Advanced Studies, New Delhi, India. ¹⁴College of Animal Science, South China Agricultural University, Guangzhou, China.

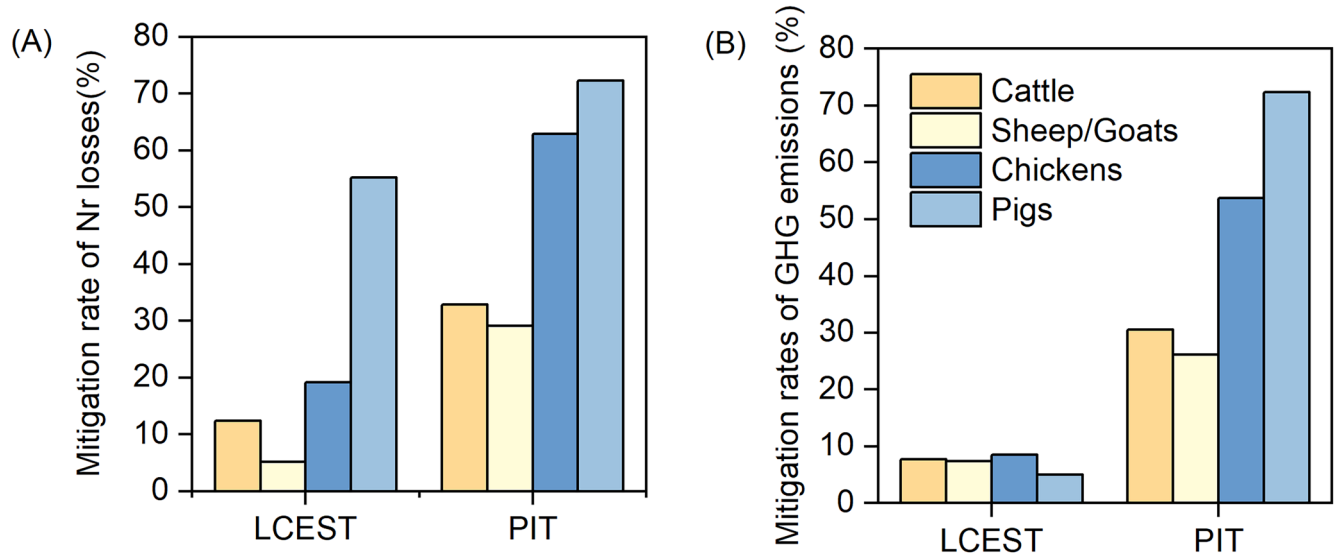
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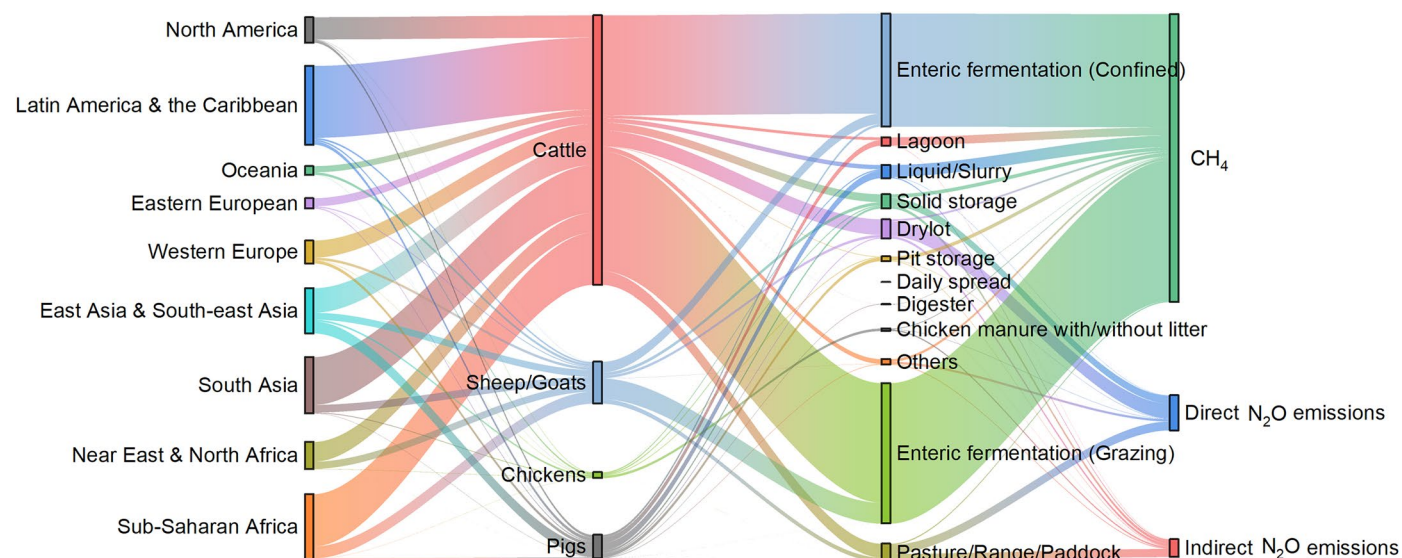
Extended Data Fig. 1 | Regional distribution of livestock reactive nitrogen losses across species, management systems, and emission pathways in 2023.

Sankey diagram showing how livestock losses flow from world regions (left nodes), through livestock species and management systems (middle nodes),

to loss pathways (right nodes). Flow width is proportional to the magnitude of losses, indicating the relative contribution of each region, species, and management system to different loss pathways. Detailed results are provided in Supplementary Text 1.

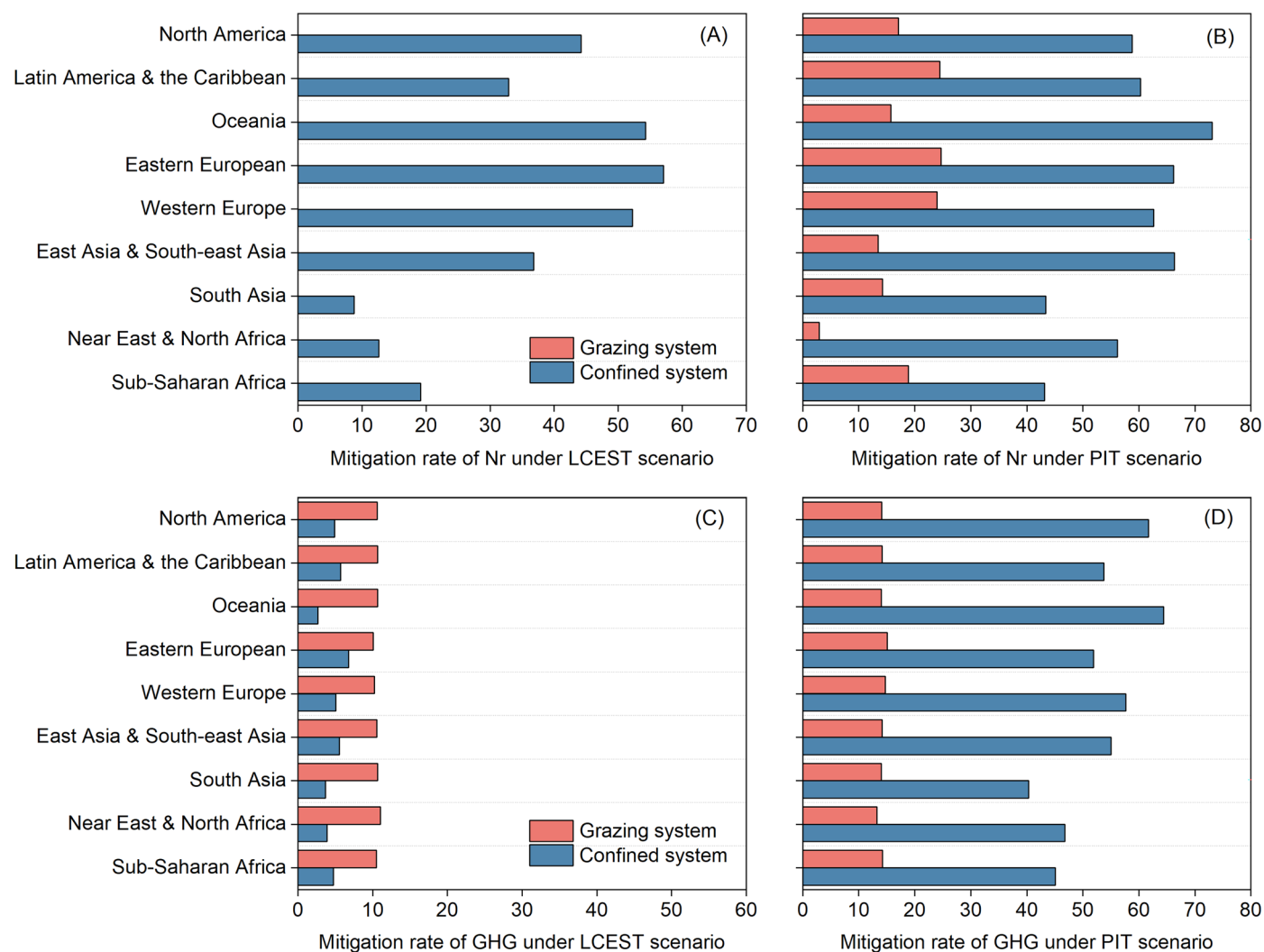


Extended Data Fig. 2 | Mitigation rates of reactive nitrogen and greenhouse gases across livestock species under different mitigation scenarios. Comparative effects of low-cost and easily scalable technologies (LCEST) and precision-integrated technologies (PIT). **A, B**, Reactive nitrogen losses (**A**) and greenhouse gas emissions (**B**).



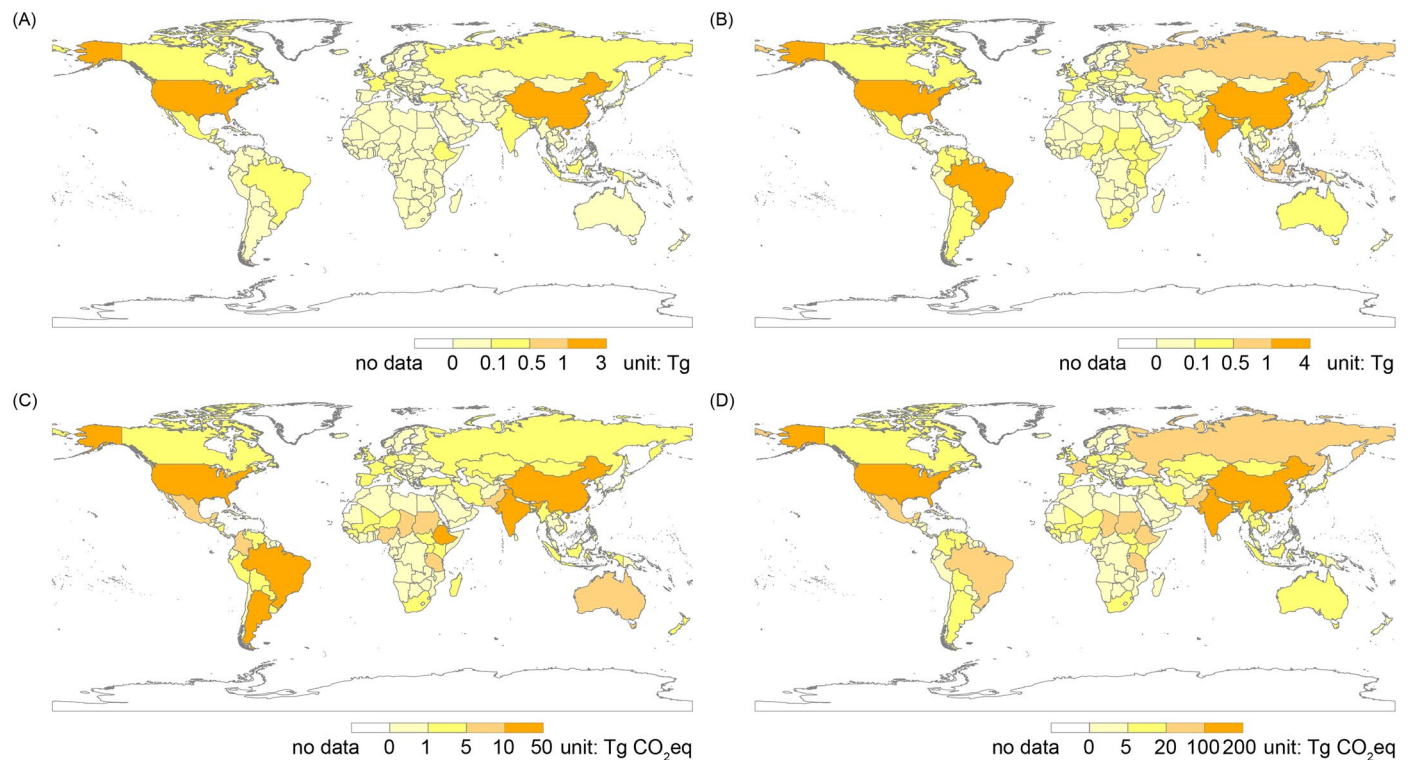
Extended Data Fig. 3 | Regional distribution of livestock greenhouse gas emissions across species, management systems, and emission pathways in 2023. Sankey diagram showing how livestock emissions flow from world regions (left nodes), through livestock species and management systems (middle nodes),

to loss pathways (right nodes). Flow width is proportional to the magnitude of emissions, indicating the relative contribution of each region, species, and management system to different emission pathways. Detailed results are provided in Supplementary Text 2.



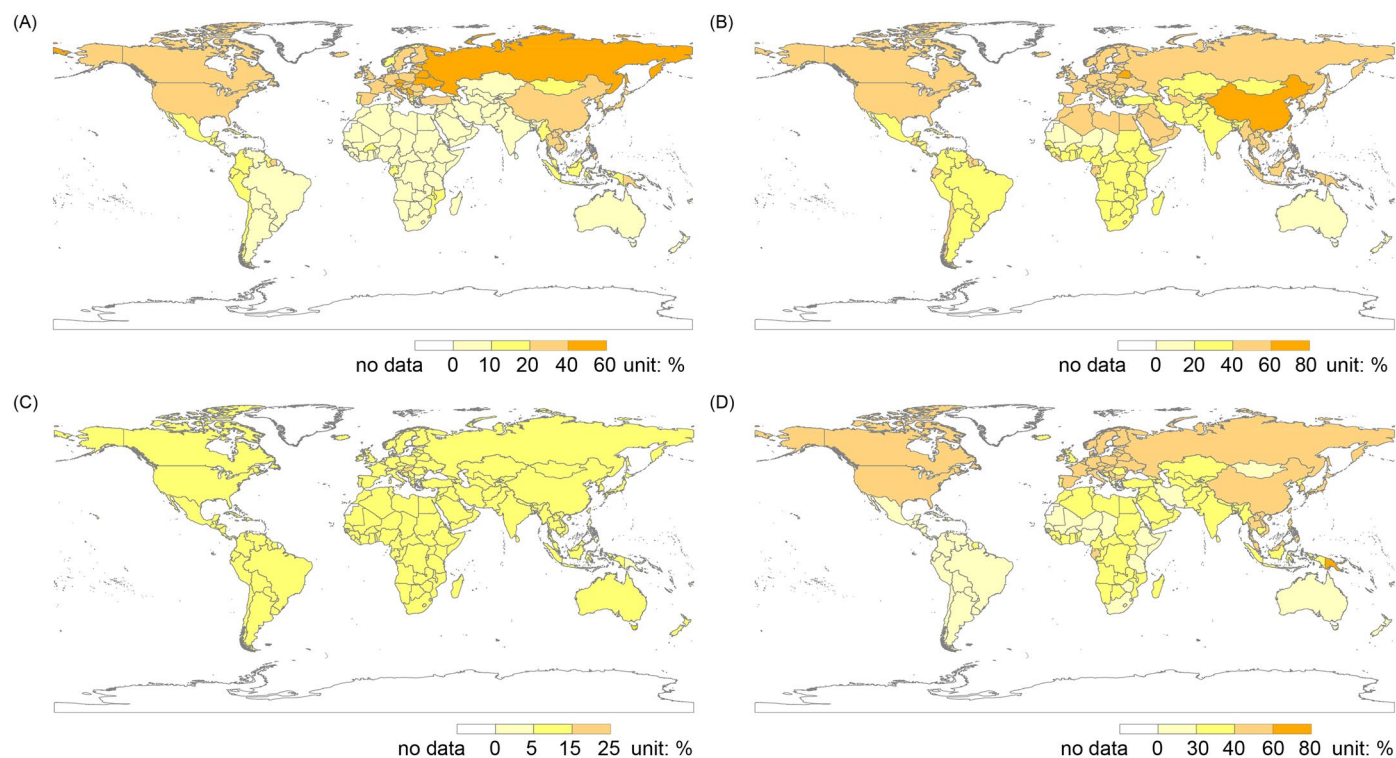
Extended Data Fig. 4 | Regional variations in mitigation rates of reactive nitrogen and greenhouse gas emissions for confined and grazing systems under different mitigation scenarios. A, B, Mitigation rates of reactive nitrogen

losses under the LCEST scenario (A) and the PIT scenario (B). C, D, Mitigation rates of greenhouse gas emissions under LCEST (C) and PIT (D), differentiated by confined and grazing livestock systems.



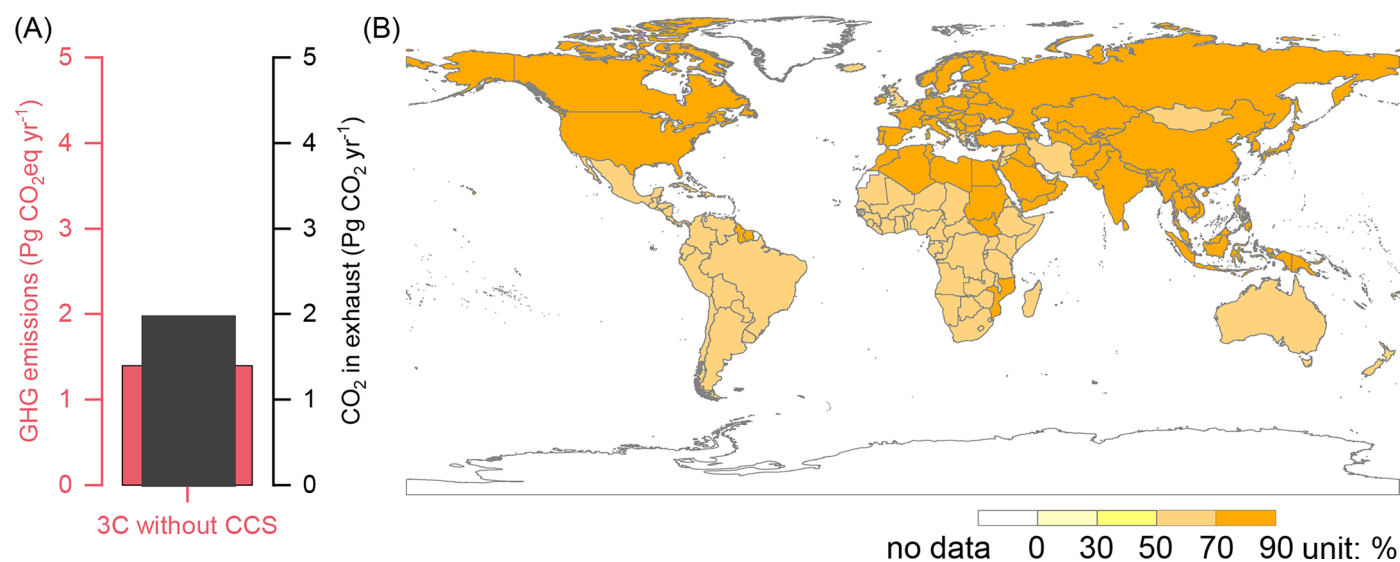
Extended Data Fig. 5 | Spatial variations in livestock reactive nitrogen and greenhouse gas mitigation potentials under different mitigation scenarios. **A, B,** Mitigation potential of reactive nitrogen losses under LCEST (**A**) and PIT (**B**). **C, D,** Mitigation potential of greenhouse gas emissions under LCEST (**C**) and PIT

(**D**). Basemap administrative boundary data from the Resource and Environment Science Data Platform, Chinese Academy of Sciences (<https://www.resdc.cn/data.aspx?DATAID=205>).



Extended Data Fig. 6 | Spatial variations in mitigation rates of livestock reactive nitrogen and greenhouse gas emissions under different mitigation scenarios. A, B, Mitigation rates of reactive nitrogen losses under the LCEST scenario (A) and the PIT scenario (B). **C, D,** Mitigation rates of greenhouse gas

emissions under LCEST (C) and PIT (D). Basemap administrative boundary data from the Resource and Environment Science Data Platform, Chinese Academy of Sciences (<https://www.resdc.cn/data.aspx?DATAID=205>).



Extended Data Fig. 7 | Greenhouse gas emissions under 3C scenarios exclude CCS technologies in 2050. A, Greenhouse gas and CO₂ emissions; **B**, Regional variation in greenhouse gas reduction rates. Basemap administrative boundary data in **b** from the Resource and Environment Science Data Platform, Chinese Academy of Sciences (<https://www.resdc.cn/data.aspx?DATAID=205>).

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- For clinical datasets or third party data, please ensure that the statement adheres to our [policy](#)

The data supporting the findings of this study are available within the article and the Supplementary Information. Detailed data sources used for the meta-analysis are listed in Supplementary Tables S9 and S15. The coefficients used in the mitigation-effect analysis are provided in Supplementary Tables S2, S4 and S7. Source

data are provided with this paper: Animal population data by country were primarily sourced from FAOSTAT 2023. Future projections to 2050 were derived from the GLOBIOM model under the Shared Socioeconomic Pathway 2 (SSP2) scenario. Annual nitrogen excretion rate was primarily obtained from the IPCC 2019 Refinement, with country-specific values for major producers calibrated using national GHG inventories submitted to the UNFCCC (2022) (<https://unfccc.int/ghg-inventories-annex-i-parties/2022>) and peer-reviewed literature. Supplementary country-specific manure management system allocation data were sourced from the IPCC 2019 Refinement (Tables 10A.6–10A.9), with country-specific adjustments applied where available using national inventory submissions and supplementary literature. Emission factors of reactive nitrogen losses and greenhouse gas emissions were primarily sourced from the IPCC 2019 Refinement (Tables 10.21–10.22; Tables 11.3, 8A.1) and the GLEAM 3.0 model (<https://www.fao.org/gleam/en/>).

Research involving human participants, their data, or biological material

Policy information about studies with [human participants or human data](#). See also policy information about [sex, gender \(identity/presentation\), and sexual orientation](#) and [race, ethnicity and racism](#).

Reporting on sex and gender

Reporting on race, ethnicity, or other socially relevant groupings

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Ethics oversight

Note that full information on the approval of the study protocol must also be provided in the manuscript.

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For a reference copy of the document with all sections, see [nature.com/documents/nr-reporting-summary-flat.pdf](https://www.nature.com/documents/nr-reporting-summary-flat.pdf)

Ecological, evolutionary & environmental sciences study design

All studies must disclose on these points even when the disclosure is negative.

Study description	Simultaneous mitigation of reactive nitrogen (Nr) and greenhouse gas (GHG) emissions in livestock systems is a critical challenge for sustainable food production. Here we conduct a meta-analysis of over 3,000 empirical observations to assess the mitigation performance of integrated technologies within livestock farms. Mitigation options are categorized into low-cost and easily scalable technologies (LCEST) and precision-integrated technologies (PIT), with the latter performing best by reducing Nr and non-CO ₂ GHG emissions by two fifths and a third, respectively. High efficacy hotspots emerged in North America, Europe, and East & Southeast Asia. Beyond current practices, a transition towards an integrated management framework centered on enclosed manure treatment systems offers further reductions of over half for Nr and two thirds for non-CO ₂ GHGs by 2050. Meeting net-zero targets will require sustained capture of almost three quarters of CO ₂ from these systems. These findings highlight the critical role of integrated approaches for advancing sustainable and climate-resilient livestock production systems.
Research sample	This study developed a comprehensive abatement technology database using meta-analytical methods, incorporating over 3,000 observations. The impact of global integration of existing technologies on livestock reactive nitrogen and greenhouse gas emissions were analyzed.
Sampling strategy	(1) four major livestock categories—cattle, sheep/goats, chickens, and swine; (2) direct and indirect non-CO ₂ greenhouse gas indicators, such as enteric methane, manure methane, nitrous oxide, nitrogen leaching, and ammonia; (3) The manure management systems assessed for these livestock included lagoon, liquid/slurry, solid storage, drylot, daily spread, digester, pit storage, chicken manure with or without litter, and others.
Data collection	Peer-reviewed literature was systematically collected using electronic databases including Web of Science and Google Scholar. Animal population data by country were primarily sourced from FAOSTAT 2023. Future projections to 2050 were derived from the GLOBIOM model under the Shared Socioeconomic Pathway 2 (SSP2) scenario. Annual nitrogen excretion rate was primarily obtained from the IPCC 2019 Refinement, with country-specific values for major producers calibrated using national GHG inventories submitted to the UNFCCC (2022) (https://unfccc.int/ghg-inventories-annex-i-parties/2022) and peer-reviewed literature. Supplementary country-specific manure management system allocation data were sourced from the IPCC 2019 Refinement (Tables 10A.6–10A.9), with country-specific adjustments applied where available using national inventory submissions and supplementary literature. Emission factors of reactive nitrogen losses and greenhouse gas emissions were primarily sourced from the IPCC 2019 Refinement (Tables 10.21–10.22; Tables 11.3, 8A.1) and the GLEAM 3.0 model (https://www.fao.org/gleam/en/).
Timing and spatial scale	Reactive nitrogen and greenhouse gas emissions from livestock system in 2023 were estimated at the global level. The effect of mitigation scenarios on global reactive nitrogen and greenhouse gas emissions and N fertilizer demand were assessed.

Data exclusions	Does not apply.
Reproducibility	Does not apply.
Randomization	Does not apply.
Blinding	Does not apply.

Did the study involve field work? ☐ Yes ☒ No

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Plants

Seed stocks	Does not apply.
Novel plant genotypes	Does not apply.
Authentication	Does not apply.